IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-36

Modulation Strategies for PWM Full Bridge Converter-II

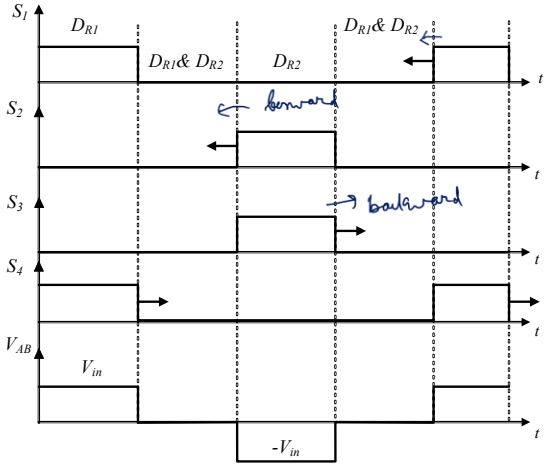
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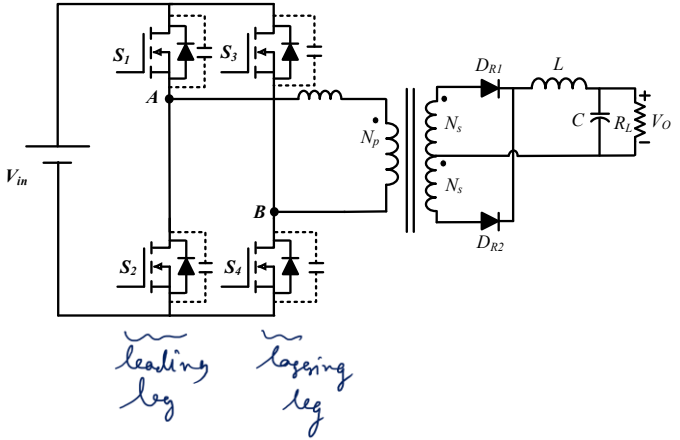


Recap

- Type-1: Diagonal switches turns-off at the same time: Modulation method 1 to 3
- Type-2: Diagonal switches turns-off at different time: Modulation method 4 to 9

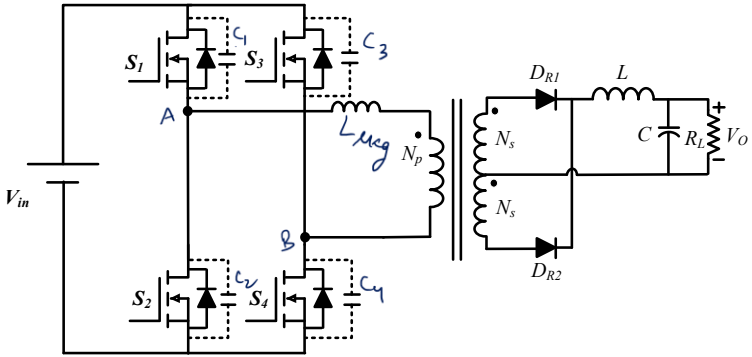


$V_o = nV_{in} \cdot D$



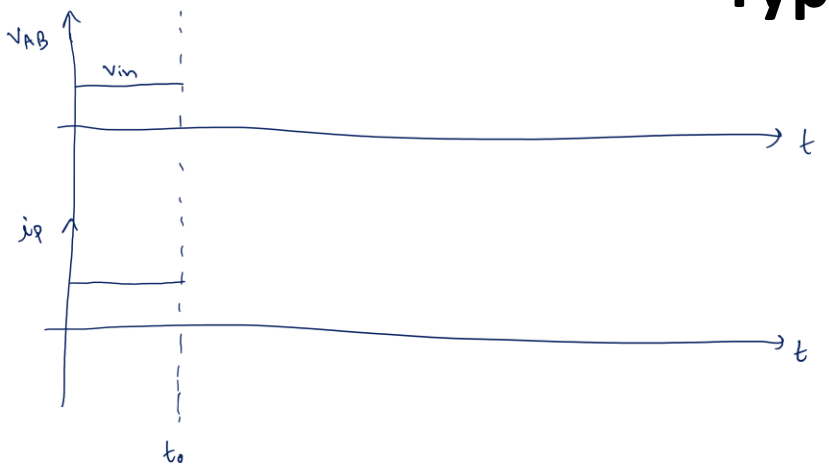
Type-1 PWM

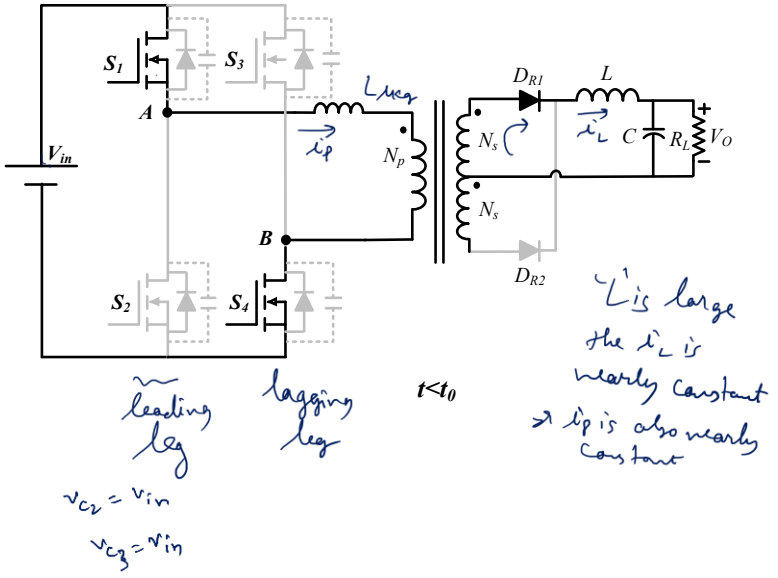
» The diagonal switches (S_1, S_4 & S_2, S_3) are turn-off at the same time.



Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-1 PWM

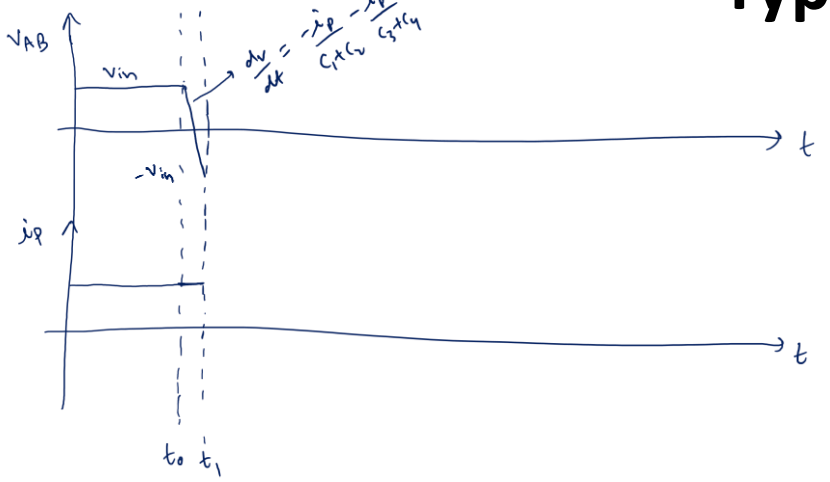


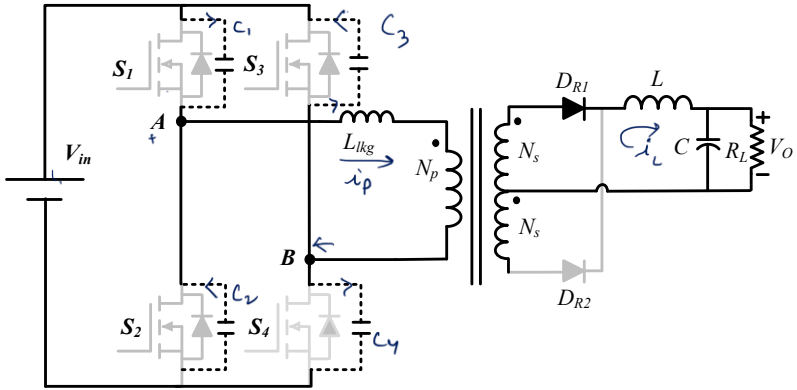


- S_1 and S_4 switches are in conduction

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-1 PWM





$t_0 < t < t_1$

C_2 discharge $\Rightarrow v_{C2} \downarrow$ from V_{in}

C_1 charges $\Rightarrow v_{C1} \uparrow$ from 0

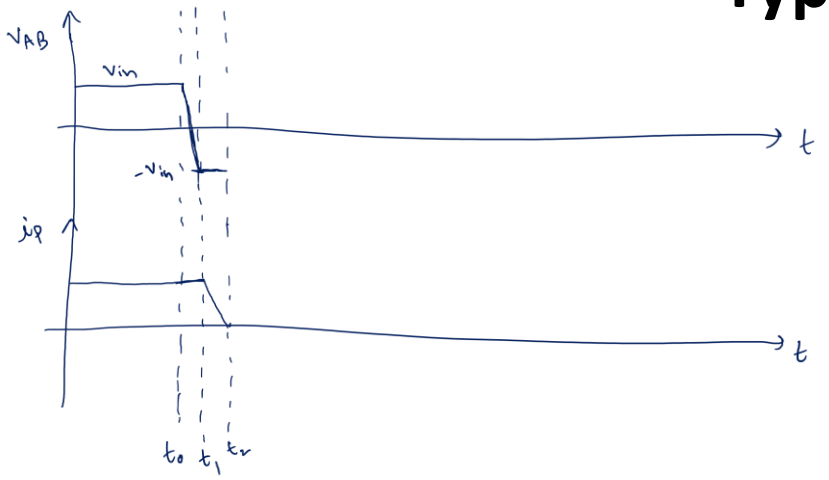
C_3 discharges $\Rightarrow v_{C3} \downarrow$ from V_{in}

C_4 charges $\Rightarrow v_{C4} \uparrow$ from 0

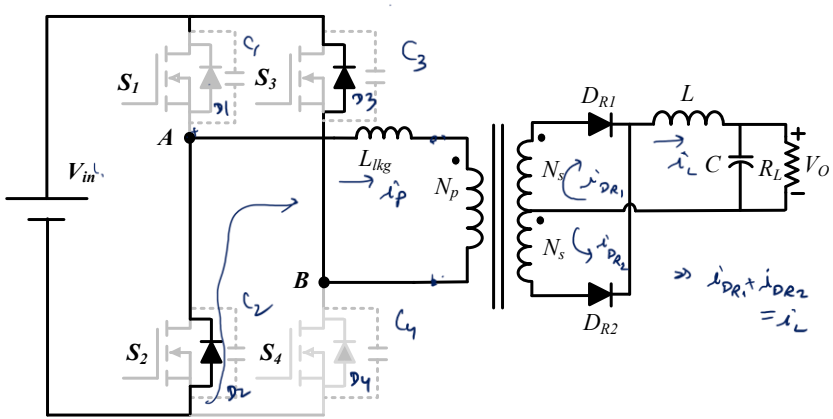
- S_1 and S_4 switches are turned-off with zero voltage
- The V_{AB} are changing with the slope defined by the output capacitance

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-1 PWM



$-V_{in}$ is appearing across the L_{lkg}



$v_{C_2} = 0$, $v_{C_3} = 0$

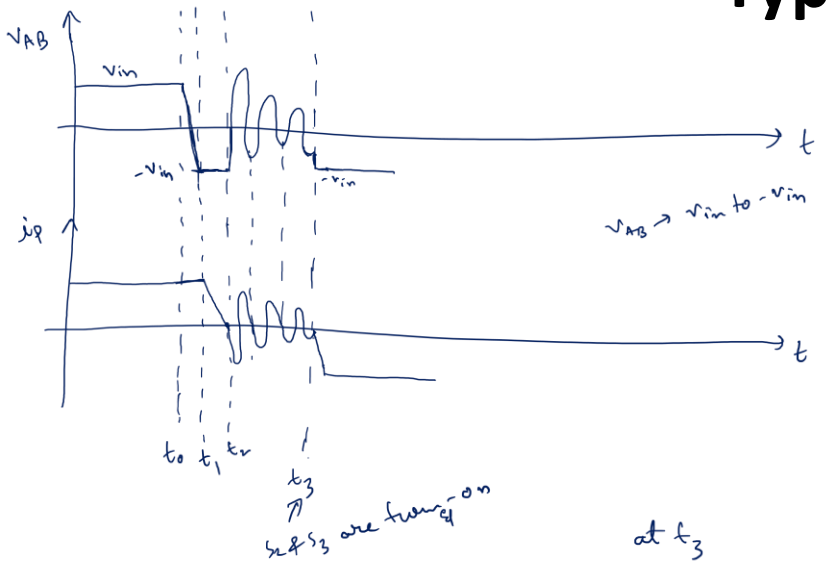
D_2 & D_3 is in conduction

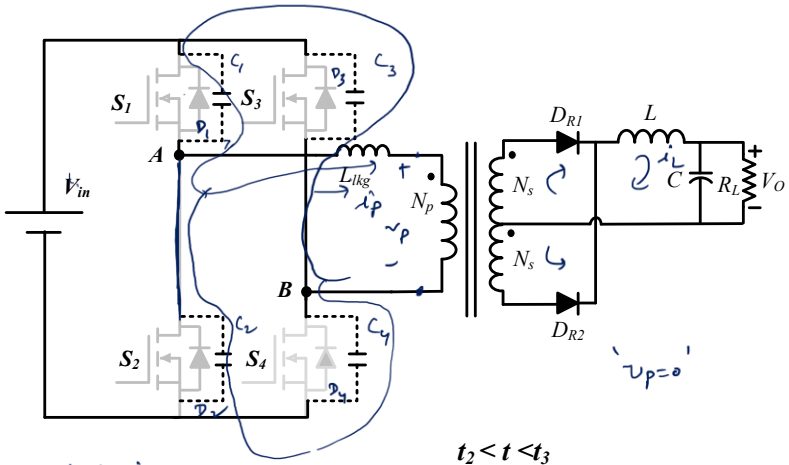
$t_1 < t < t_2$

- S_1 and S_4 switches are turned-off with zero voltage
- Transformer current goes to zero

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-1 PWM





D_2 & D_3 ceases to be in conduction

at t_3

⇒ the switch S_2 & S_3 turn on

- L_{lkg} resonates with the C_1, C_2, C_3, C_4
- S_2 , S_3 may goes to hard turn-on → the S_2 & S_3 turns on with non-zero voltage across them

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-1 PWM

Summary

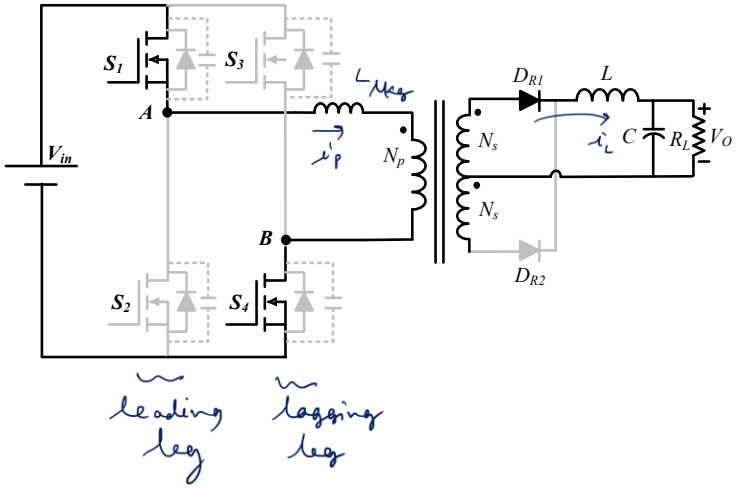
- S_1, S_2, S_3, S_4 are turned-off with zero voltage (soft-turn-off)
- S_1, S_2, S_3, S_4 are turned-on with non-zero voltage (hard turn-on)
  - May or may not

the diagonal switches turn-off simultaneously

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-2 PWM

→ turn-off of diagonal switches at different time



$V_{AB} = V_{in}$

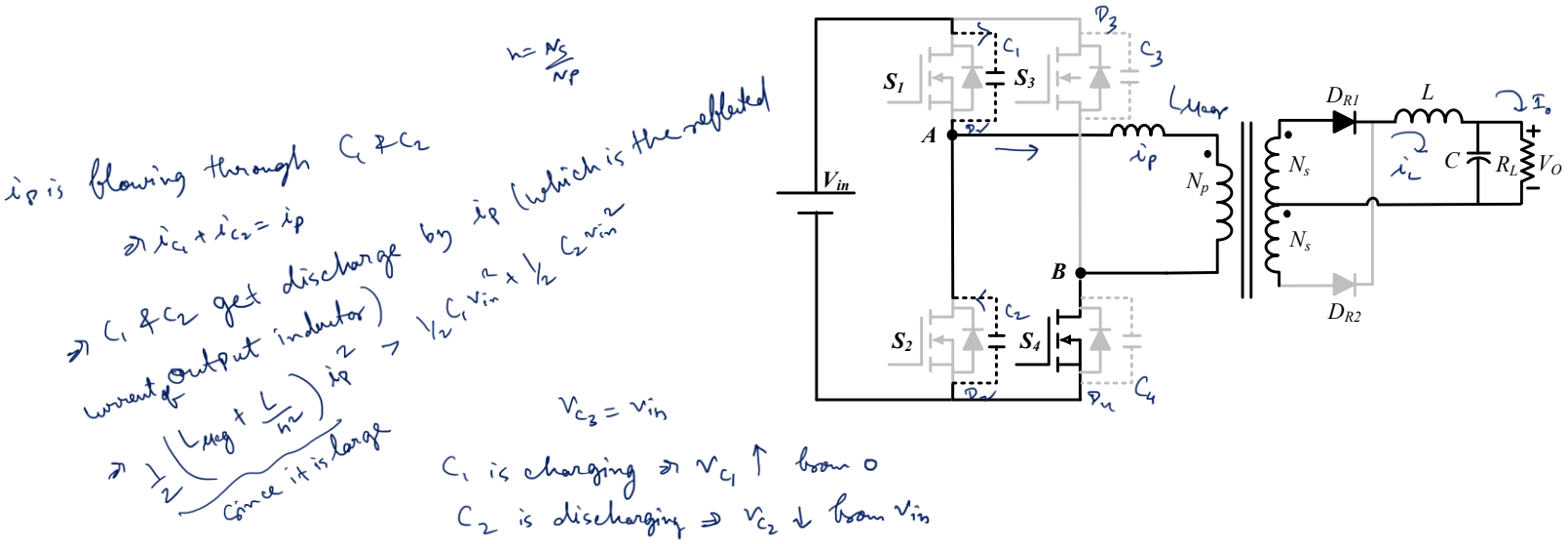
$V_{C2} = V_{in}$

$V_{C3} = V_{in}$

- S_1 and S_4 switches are in conduction

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

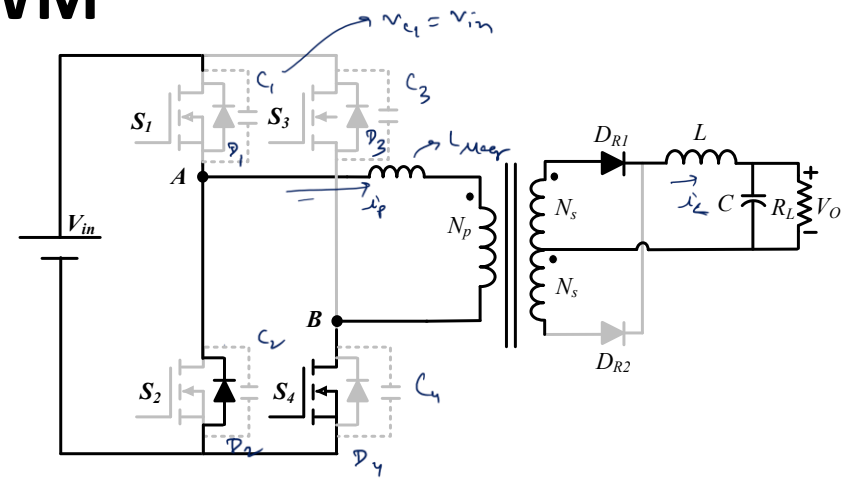
Type-2 PWM



- S_1 is turned-off and S_4 continues conduction

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

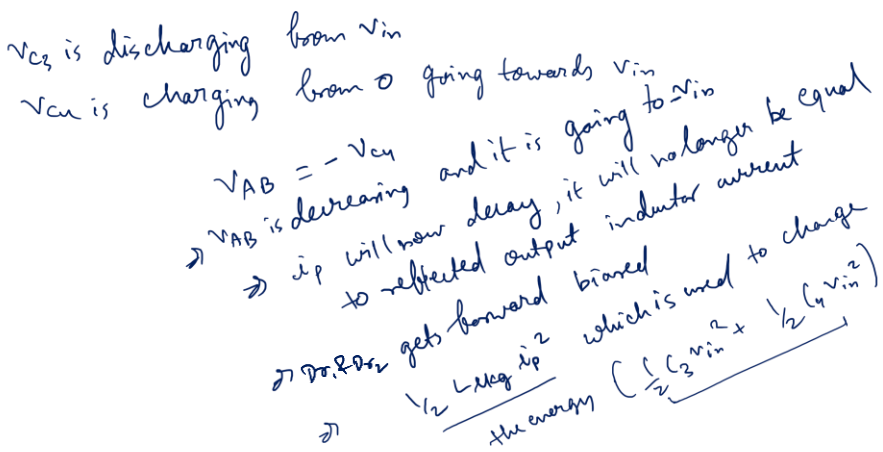
Type-2 PWM

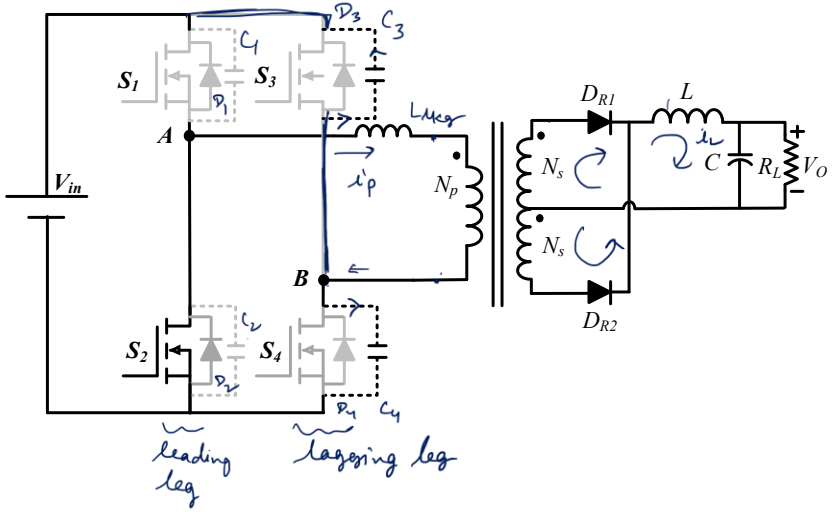


- Zero state
- $V_{AB} = 0$
- Transformer current either kept constant or current forced to zero
- S_2 turns-on with zero voltage

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Type-2 PWM





- $V_{AB} = 0$ to $-V_{in}$
- S_4 turns-off with limited dv/dt at the output due to C_3 and C_4
- S_3 may or may not go to zero voltage turn-on – due to insufficient energy due to i_p in L_{lkg}
- ZVS for lagging leg is difficult if the transformer current and L_{lkg} is not sufficient to discharge the C_3 to zero and charge the C_4 to V_{in}

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Summary

Type-2

→ The leading leg switches (S_1 & S_2) goes to ZVS during turn-on (Soft turn-on)

→ The lagging leg switches (S_3, S_4) may or may not go to zvs turn-on (hard turn-on)

→ light loading, if the Lekg is very small

Type-1 (diagonal switches are turned-off at the same time)

S_1, S_2, S_3, S_4 turn-off with zvs →

turn-on without zvs (hard turn-on) →

Type-2

leading leg goes to zvs turn-on

lagging leg may or may not goto zvs turn-on

S_1, S_2, S_3, S_4 → turn-off with zvs or limited dv/dt at the output of half-bridges

Thank You